The forward model, and three sweeps that test it

Fluo-BPM, at the optics of `confocal_emf3.czi`*. 2026-08-05*

1. The model, in one page

Propagation. Maxwell's equations for a monochromatic scalar field in a non-magnetic medium reduce to the Helmholtz equation

$$\nabla^2 U(\vec{r}) + k(\vec{r})^2 U(\vec{r}) = 0, \qquad k(\vec{r}) = \frac{2\pi n(\vec{r})}{\lambda}, \tag{1}$$

which keeps the full three-dimensional index distribution $n(\vec{r})$ but drops polarization. Splitting the index into a background n_0 and a deviation $\delta n = n - n_0$, and keeping only forward propagation, the volume is treated as P slices of thickness Δz, each carrying a phase map $\phi_k(x, y) = 2\pi\delta n_k(x, y)\Delta z/\lambda$. One slice to the next is then

$$U_{k+1} = \mathcal{F}^{-1}\Big[\, \mathcal{F}\big[\, U_k\, e^{i\phi_k} \,\big] \odot \tilde{H}_{\Delta z} \Big], \qquad \tilde{H}_{\Delta z}(\mu_x, \mu_y) = e^{\,i2\pi\Delta z\sqrt{n_0^2/\lambda^2 - \mu_x^2 - \mu_y^2}}, \tag{2}$$

a phase multiplication for what the light passes through, then a Fresnel propagation for how it diffracts on the way to the next slice. $\odot$ is point-wise multiplication. Equation (2) is the whole propagation engine; everything else is what is put into it and what is read out of it.

Fluorescent sources. Emitters add a source term, and their incoherence is the statement that different points are uncorrelated,

$$E\big[s^*(\vec{r}\,')\, s(\vec{r})\big] = \delta(\vec{r}\,' - \vec{r})\, F(\vec{r}), \tag{3}$$

with F the fluorophore distribution. The measured intensity is then $I(\vec{r}\,') = \int |G(\vec{r}\,', \vec{r})|^2 F(\vec{r})\, d\vec{r}$, where G is the Green function that Eq. (2) implements. That integral has no closed form, because it needs $|G|^2$ rather than G. It can, however, be sampled: drawing a random phase $\psi \sim \mathcal{U}[0, 2\pi]$ per emitter and propagating the *coherent* field $\sqrt{F}\, e^{i\psi}$ gives

$$I = E_\psi\Big[\, \big|\, G\left(\sqrt{F} \odot e^{i\psi}\right) \big|^2 \,\Big], \tag{4}$$

since a diagonal source covariance is exactly what that draw produces: $\Gamma_U = G\,\Gamma_S\, G^H$ with $\Gamma_S = \mathrm{diag}(F)$ gives $\Gamma_{U,ii} = \sum_j |G_{ij}|^2 F_j = I_i$. *Incoherent light is thus simulated as an average of coherent propagations.* In the slice recursion, with $w = 1 \dots W$ indexing the draws,

$$U_{k+1,w} = \mathcal{F}^{-1}\Bigg[\, \mathcal{F}\Big[\, U_{k,w}\, e^{i\phi_k} \;+\; \mathcal{F}^{-1}\big[\, \mathcal{F}[\, \sqrt{F_k}\, e^{i\psi_{k,w}} \,] \odot C^{-1} \big] \,\Big] \odot \tilde{H}_{\Delta z} \Bigg], \tag{5}$$

where $C = \sqrt{n_0^2/\lambda^2 - \mu_x^2 - \mu_y^2}$ and $C^{-1} \propto \cos(\theta)^{-1}$ weights emission away from the optical axis.

Detection and the average. The field leaving the last slice is filtered by the pupil and back-propagated to each focal plane:

$$I_{k,w} = \Big|\, \mathcal{F}^{-1}\big[\, \mathcal{F}[U_{P,w}] \odot P_{\mathrm{circ}} \odot e^{i\Gamma} \odot \tilde{H}^*_{(P-k+1)\Delta z} \,\big] \Big|^2, \qquad I_k = \frac{1}{W}\sum_{w=1}^{W} I_{k,w}, \tag{6}$$

with $P_{\mathrm{circ}} = 1$ for $\sqrt{\mu_x^2 + \mu_y^2} < \mathrm{NA}/\lambda$ and zero outside, and $e^{i\Gamma}$ an optional known aberration (a weighted sum of Zernike polynomials).

Two consequences that bit in practice. First, $\psi_{k,w}$ carries the index k: the phase must be redrawn *per slice*, not once per draw. Sharing one screen down the volume makes every emitter in an (x, y) column mutually coherent, so a cell's slices interfere on axis and Fresnel rings appear at the centre of every cell, identically in every draw — averaging cannot remove them. Restoring per-slice phases removed the rings and doubled the mean in-cell signal, which the interference had been redistributing. Second, P_{circ} is only a filter if $\mathrm{NA}/\lambda < 1/(2\,dx)$; sampled coarser than that, the pupil is wider than the represented k-space, nothing is filtered, and the model has no optical sectioning at all.

2. Sweep 1: the Monte-Carlo average

Equation (4) is an expectation, and a finite W leaves residue that looks exactly like detector noise. With the detector switched off entirely, the stack still fluctuates by 9.5 % of the local mean at $W = 30$, falling to 2.0 % at $W = 1920$ (Figure 1). The decay is close to but slower than the $1/\sqrt{W}$ that independent draws would give: fitted over this range it goes as $W^{-0.38}$, so 64 times the draws bought a factor 4.8 rather than 8. Draws are not fully independent here — the same phantom and the same haze are common to all of them — so the ideal law is a bound, not a prediction. Grain in a folder whose name says no measurement noise is this. The sweeps below run at $W = 240$ and $W = 1920$ respectively.

3. Sweep 2: refractive index

Five arms, index contrast 0 to 0.02, 384 µm deep, no measurement noise, same cells and seed throughout. The objective sits past the exit face, so light from the far plane crosses the whole sample and must arrive degraded while the near plane does not — and $dn = 0$ is the control that must come out flat.

arm	contrast near objective	contrast far side	lost across the depth
dn = 0	0.060	0.046	24%
dn = 0.0025	0.059	0.037	37%
dn = 0.005	0.058	0.023	61%
dn = 0.010	0.054	0.004	93%
dn = 0.020	0.045	-0.000	101%

At $dn = 0.02$ the far side has no cell contrast left at all while the near side keeps three quarters of the index-matched value (Figures 2 and 3). Two notes on measuring this. Gradient energy — the obvious sharpness metric — gives a near/far ratio of 1.00 to 1.01 across the entire sweep and misses the effect completely, because in a stack this dense it is dominated by fine haze texture that survives; what refraction destroys is the difference between a cell and the gap beside it. And ratios are the wrong summary once the far-side contrast crosses zero: near/far runs to +13.8 and then to −109.

Integrated brightness is also the wrong observable: a sphere with $dn = 0.02$ deflects light by about 2°, and NA 1.0 in water accepts a half-angle of 48.7°, so refraction redirects light *within* the collection cone rather than out of it. An earlier claim in `DATASET.md` of a 30 % depth attenuation came from runs with the shared-phase bug above and does not survive.

4. Sweep 3: PSF elongation

Lowering NA stretches the PSF axially far faster than laterally — lateral FWHM $\sim 0.5\lambda/\mathrm{NA}$, axial $\sim \lambda/(n - \sqrt{n^2 - \mathrm{NA}^2})$ — so cells should stay about the same size in xy and stretch in z. Five arms, NA 1.0 down to 0.3, no refraction and no measurement noise, $W = 1920$.

NA	apparent lateral extent	apparent axial extent	axial / lateral
1.0	8.03 µm	14.00 µm	1.7
0.8	7.90 µm	17.00 µm	2.2
0.6	8.03 µm	22.00 µm	2.7
0.4	8.16 µm	31.00 µm	3.8
0.3	8.55 µm	36.50 µm	4.3

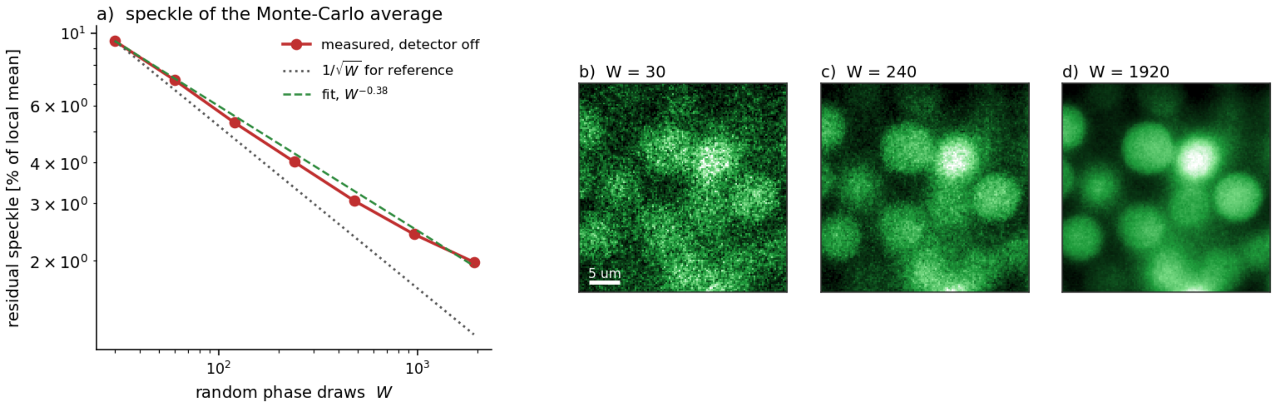

Figure 1: *

Figure 1. a) Residual speckle of the Monte-Carlo average against the number of random phase draws, detector off, with the fitted power law and $1/\sqrt{W}$ for reference. b–d) The same plane at three values of W.

Figure 4 shows it directly: xy sections barely change while xz sections stretch into cigars.

Caveat throughout. The reference is a confocal LSM 800 with a 2.54 AU pinhole. This model detects widefield — every slice reaches the detector — so these stacks carry more out-of-focus haze than the real data. Equation (5) supports the light-sheet case (excite one slice at a time), but a pinhole is not a parameter change.

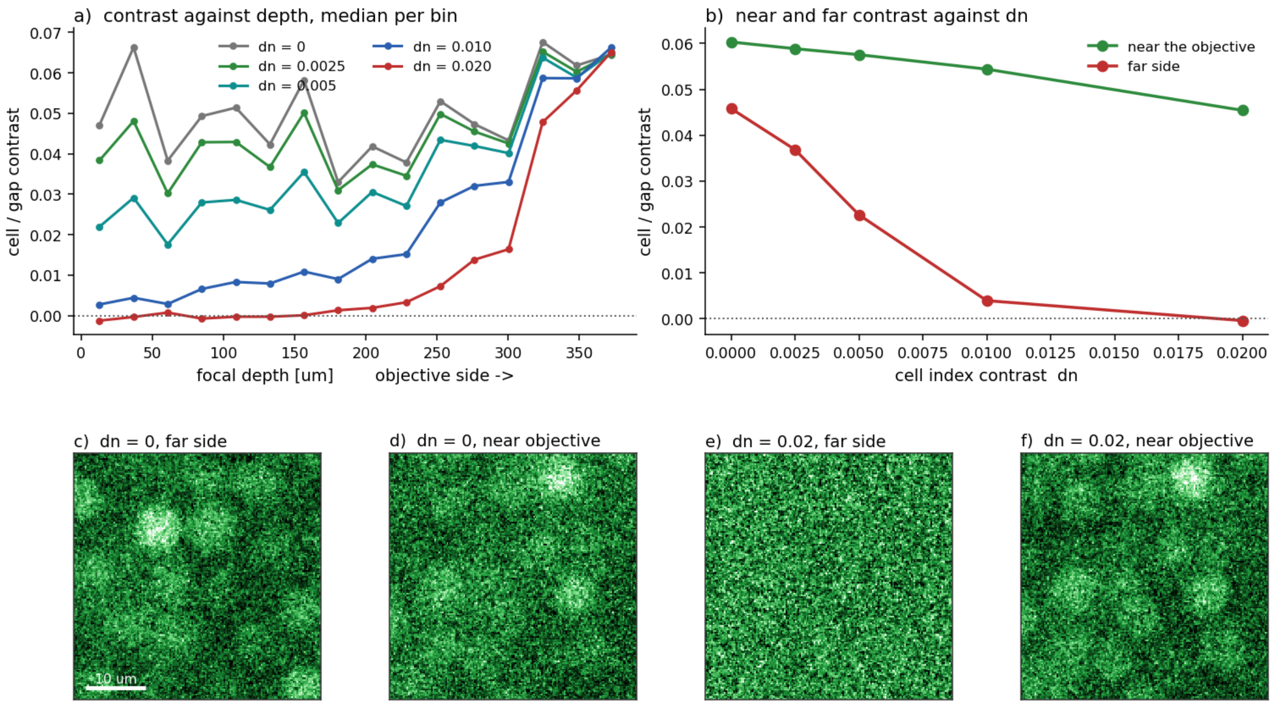

Figure 2: *

Figure 2. Index sweep, 384 µm deep, no measurement noise. a) Cell/gap contrast against focal depth, median per depth bin. b) The same near the objective and on the far side, against dn; dotted line is zero contrast. c–f) The stacks themselves: the index-matched control looks the same at both ends, while at $dn = 0.02$ the far side has lost its cells. These arms run at $W = 240$, so the visible grain is the Monte-Carlo residue of Figure 1 (2.8 %), not the detector, which is off.

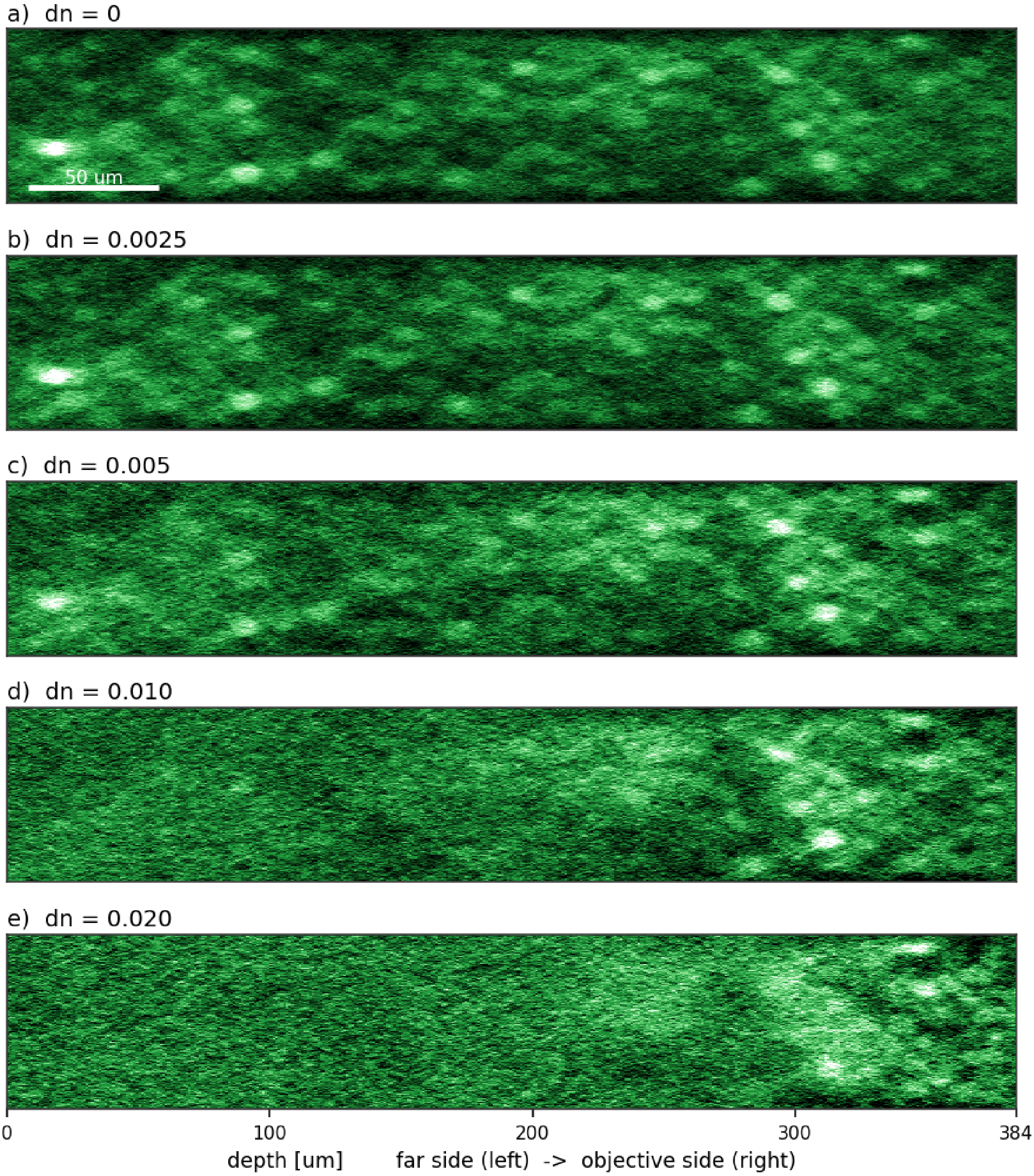

Figure 3: *

Figure 3. The index sweep seen edge-on: an xz strip through each arm, the full 384 µm of depth on the horizontal axis, far side on the left and objective on the right. Identical crop, scale and display range in every panel; averaged over 1.5 µm in y for display, equally in all arms. The cells fade from the left as the index contrast rises, while the right-hand end holds.

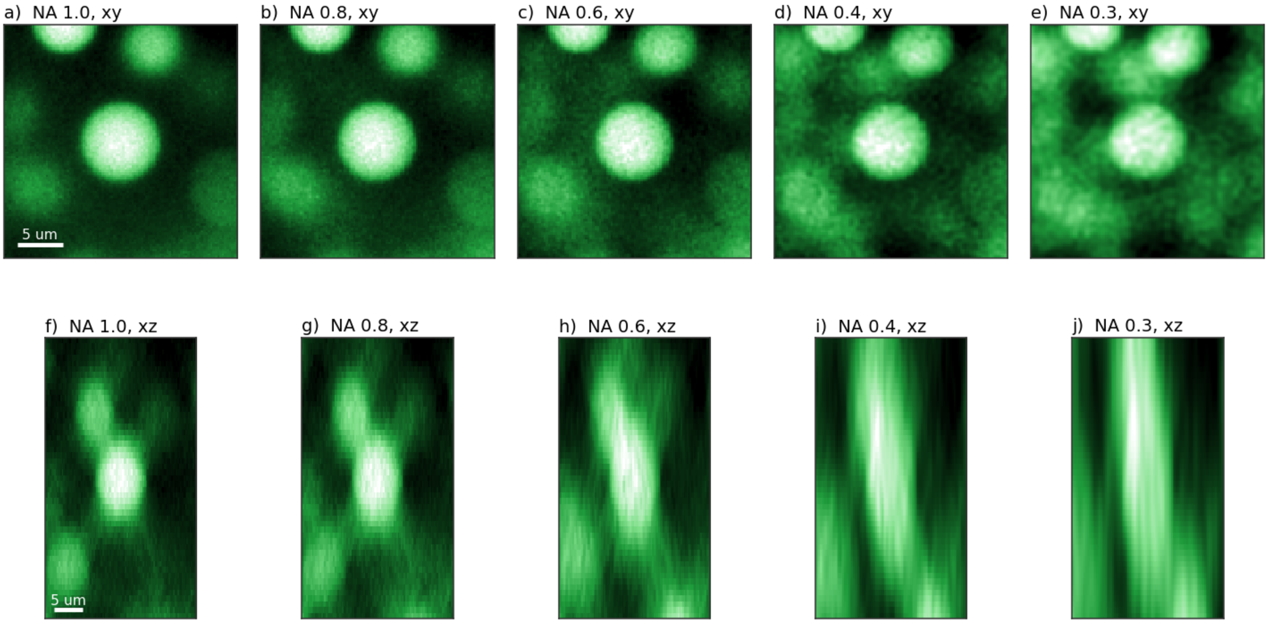

Figure 4: *

Figure 4. NA sweep, all panels the same isolated cell (the arms share a seed) at the same physical scale. Top row xy, bottom row xz. Lower NA leaves xy nearly unchanged and stretches z.

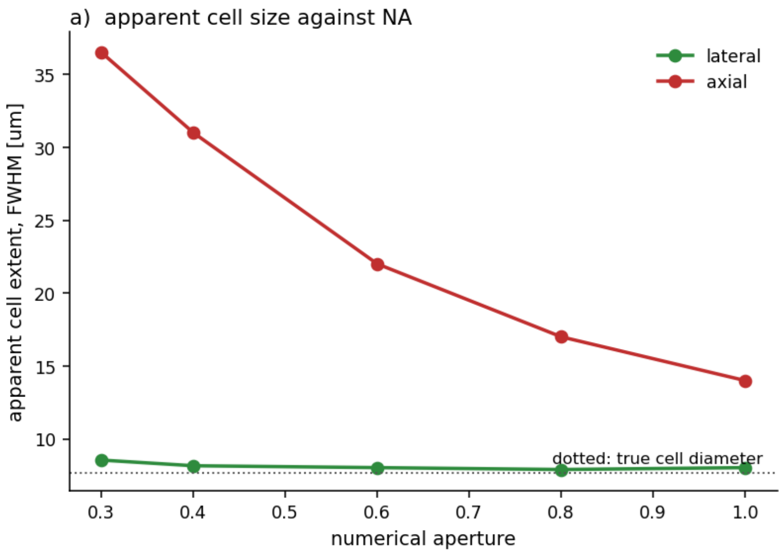

Figure 5: *

Figure 5. Apparent cell extent, measured at half maximum through each cell's own centre, against NA.